



MANIPAL ACADEMY OF HIGHER EDUCATION

Manipal Institute of Technology, Bengaluru

BTech VI Semester

End Semester Examination - April / May -2024

SOFTWARE VERIFICATION AND VALIDATION [IT-3223]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) What makes a good software test engineer? Give the principles of software testing. (5)
 - A)
 - B) Why does software have bugs? Illustrate the reasons with an example. (5)
- 2) Describe the impact of engineer's soft skills for performing verification techniques. (5)
 - A)
 - B) How defect severity and priority are determined during software testing. Provide examples to justify your explanation. (5)
- 3) Compare software inspection with testing in terms of defect detection capabilities, cost-effectiveness, and timing in the software development lifecycle. Why are inspections considered more effective at detecting certain types of defects? (5)
 - A)
 - B) What is equivalence class partitioning? Design Strong normal equivalence class test cases for the NextDate function assuming necessary data. (5)
- 4) Derive the boundary value test cases for the Simple Automated Teller Machine assuming necessary data. (5)
 - A)
 - B) Draw the control flow graph and Compute the cyclomatic complexity for the given program. (5)

1. Input (a, b, c);
2. $D \leftarrow b*b-4*a*c$;
3. If($d>0$) then
4. $R \leftarrow 2$;
5. else if ($d=0$) then
6. $r \leftarrow 1$;
7. else if ($d<0$) then
8. $r \leftarrow 0$;
9. output (r);

- 5) Illustrate the factors that are considered for identifying resource requirements for planning a testing project? (5)
- A)
- B) Explain the need of test reporting. Explain the different types of test reports. (5)

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